Speeding up reactions

Answers

Boiling the billy

1. Making a pot of tea at a camp fire is known in Australia as 'boiling the billy'. The camper wants to boil the water faster. An increase in the rate of wood burning will deliver heat to the water faster.



(a) One method of increasing the combustion of the wood is to add more reactants. How is the paper fanning doing this?
Adding more oxygen from the air at a greater rate than just diffusion

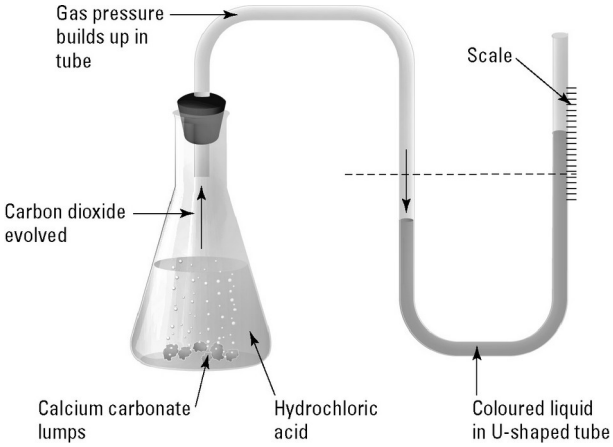
(b) Increasing the surface area of the wood also helps. How has the camper achieved this?
Cutting the wood into small sticks, which have a greater surface area

Changing the speed of the reaction

2. Calcium carbonate is the main chemical present in limestone. When hydrochloric acid is added to lumps of calcium carbonate, it fizzes and dissolves to form calcium chloride and water; gaseous carbon dioxide is also formed. The calcium chloride remains dissolved in the acid.

(a) Write a word equation for this reaction.
Calcium carbonate + hydrochloric acid → calcium chloride + water + carbon dioxide

(b) A student used the following apparatus to study the speed of the reaction.



(i) Explain how the apparatus can be used to measure how fast the calcium carbonate is dissolved in the acid.
The faster the reaction, the more rapidly the gas is evolved and the faster it depresses the liquid in the left arm of the U-tube. Liquid rises in the right arm and the scale can be used to measure the change at regular time intervals.

(ii) Compare the observations when the experiment is repeated with powdered calcium carbonate instead of lumps of calcium carbonate.
The rate will be faster. The effervescence will be greater as the powder has a greater surface area.

(iii) The student heats the flask to a higher temperature. How will the observations change compared with the experiment at room temperature?
Increasing the temperature increases the rate and so fizzing will occur faster and the change in levels in the U-tube will be faster.

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